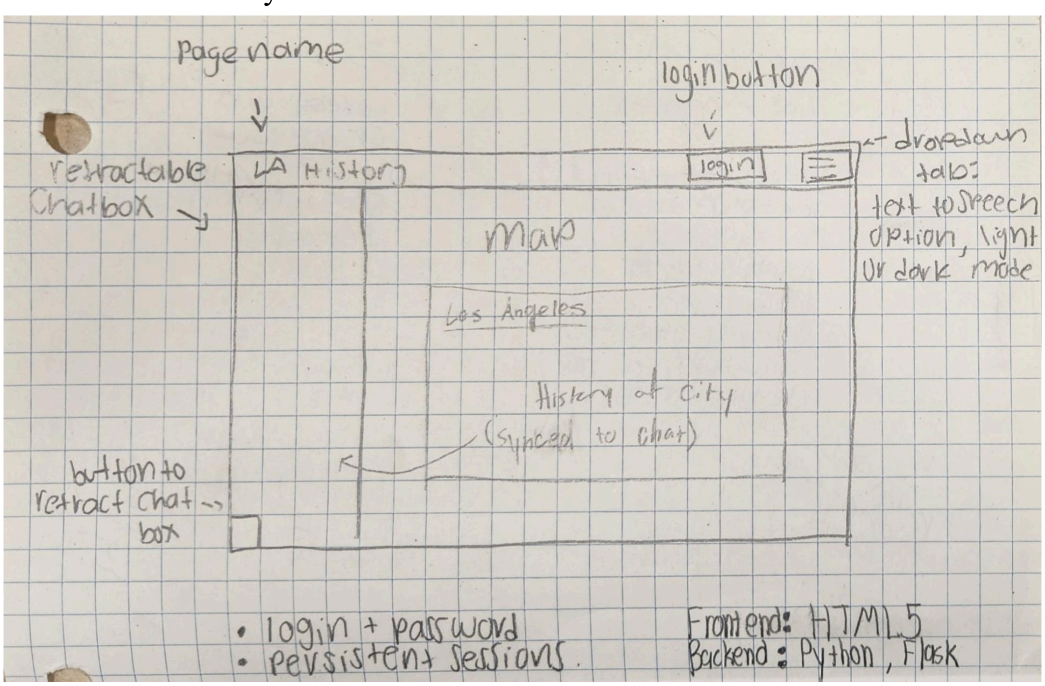
1. **Problem Statement & Learner Persona.** Who is your learner? What do they struggle with? Cite at least one source (research or practitioner literature) that supports the learning problem you are addressing.
    - Our target audience will consist of high school and college students who did not grow up in Los Angeles but want to learn more about the city. These learners face challenges in developing a contextual understanding of L.A. history and its significance due to a lack of prior exposure or local familiarity.
    - One source that supports the problem:
        - Hattan, C., Alexander, P. A., & Lupo, S. M. (2024). Leveraging What Students Know to Make Sense of Texts: What the Research Says About Prior Knowledge Activation. *Review of Educational Research*, *94*(1), 73-111.
            - Students who did not grow up in Los Angeles tend to lack familiarity with the city's neighborhoods, cultural landmarks, and local events that those who grew up there would naturally know. Because of this, they have little to connect new information to when learning about L.A. history. According to schema theory, students can only make sense of new information when they are able to connect it to knowledge they already have (Anderson & Pearson, 1984, as cited in Hattan et al., 2024).
2. **Learning Theory Rationale.** Name the theory (or theories) grounding your design. Explain how the theory maps to at least three specific features of your application.
    - Our design is grounded in Constructivism, a learning theory mentioned in the How People Learn reading that our class has been focusing on. This theory focuses on how learners actively construct knowledge through experience, reflection, and interaction rather than passively receiving information. However, our design will also feature the theories known as the zone of proximal development and transfer as guides.
        - **AI-guided hints and suggestions:** When a user becomes stuck or confused while exploring the map, the AI provides built-in support. This aligns with constructivist principles by offering “just in time” assistance that helps learners build their understanding within their **Zone of Proximal** development without directly giving away the answers.
        - **Interactive exploration:** Rather than presenting information through static text, our application will allow users to navigate historical sites, click on points of interest, and uncover narratives at their own pace. This hands-on approach will enable learners to construct knowledge through direct engagement with the content.
        - **Unlockable events for progression:** To access later historical content, users will first apply knowledge gained from earlier sections of L.A.

history. This reinforced the constructivist idea that new learning builds upon prior understanding, encouraging students to synthesize and **transfer** knowledge across contexts.

3. **AI Modality Justification.** You must use at least one LLM, VLM, or speech modality. Justify why this size of model works for your use case, and provide a cost/privacy analysis. You may need to do some research here regarding cloud providers.
    - LLM (Main) (Using Ollama)
        - This works for us because it allows the user to input text (such as by answering or asking a question) and receive personalized feedback and tutoring services. It'll be able to provide explanations, insights, feedback, and generate relevant questions that will promote user understanding and reasoning
    - TTS
        - This works for us because we are planning on having various descriptions and explanations for our application, so having the TTS option may make the experience better for certain users who wish to take advantage of the feature
    - STT (optional)
        - If we have time for it, we also plan on having a speech to text feature where the student speaks into the chatbot, the app transcribes the speech and the LLM then responds, making our application a more user friendly and accessible learning experience
4. **System Architecture Sketch.** A diagram (hand-drawn is fine) showing: frontend → backend → LLM API → any other services.
    - 

5. **Prompt Optimization Method Selection.** Choose one of the three methods described in Section 5. Define your evaluation criteria before you have written any prompts.
    - We will use Method B, which focuses more on creating prompts that guide the LLM to act as an effective tutor rather than an answer engine.
    - Our LLM prompt will *theoretically* result in responses that assess what the student has learned, asking clarifying questions to gauge comprehension before moving forward.
    - The LLM will pose analytical, evaluative, or connective questions that encourage students to think critically about historical events, causes, and consequences rather than merely recalling facts. Our prompt will constrain the LLM to avoid directly supplying answers to the users. Instead, it will offer hints, pose follow-up questions, and guide students toward discovering answers on their own, fostering deeper learning.
6. **User Testing Plan (draft)**. Who will your test participants be? How will you recruit them (min. 4 people outside your team)? What tasks will they perform?
    - Our test participants will be individuals who did not grow up in Los Angeles and/or have limited prior knowledge of the city's history. This ensures that we gather feedback from users who reflect our target audience.
    - We will recruit at least four participants from among classmates, friends, and acquaintances who meet the criteria of being non-native to Los Angeles. All participants will be outside of our immediate project team to maintain objectivity in feedback.
    - Participants will navigate the interactive map and engage with its features while performing a think-aloud protocol, verbally sharing their thoughts, reactions, questions, and any points of confusion as they progress through the experience. This approach will provide the team with valuable feedback on usability, clarity of content, and the effectiveness of the AI tutoring interactions. We will observe their session, take notes, and follow up to capture feedback.
7. **Github Repository.** Link to your team repo, set up with a README, .gitignore, and initial project structure.
    - https://github.com/AI-at-Oxy/LA-History

References

Hattan, C., Alexander, P. A., & Lupo, S. M. (2024). Leveraging What Students Know to Make Sense of Texts: What the Research Says About Prior Knowledge Activation. *Review of Educational Research*, *94*(1), 73-111.